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MGMT Promoter Methylation Classification Using 3D DenseNet and Clinical Fusion

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Certificate

This is to certify that the B.Tech Final Year Project report titled **"MGMT Promoter Methylation Classification Using 3D DenseNet and Clinical Fusion"** has been submitted by Abbas Mulla Abdul Tayyab Hakimi (Roll No.: 202251002) in partial fulfilment of the requirements for the degree of Bachelor of Technology from the Indian Institute of Information Technology Vadodara. The work presented in this report is original, carried out independently under my supervision, and has not been submitted to any other institution for the award of any degree or diploma.

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Declaration

I, **Abbas Mulla Abdul Tayyab Hakimi**, Roll No. **202251002**, pursuing B.Tech from the Indian Institute of Information Technology Vadodara, hereby declare that the work presented in this report titled **"MGMT Promoter Methylation Classification Using 3D DenseNet and Clinical Fusion"** is my own original work carried out under the supervision of **Dr.Sanjay Saxena** during the academic year 2025–26.

I further declare that:

- The work presented here has not been submitted, either in part or in full, to any other university or institution for the award of a degree or diploma.
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Abstract

One of the most aggressive and potentially fatal brain tumors, glioblastoma necessitates precise diagnosis and efficient treatment planning. The methylation state of the MGMT (O6-methylguanine-DNA methyltransferase) promoter is a crucial biomarker utilized in clinical decision-making, and it is important in predicting how well a patient will respond to chemotherapy. Traditionally, invasive methods like biopsy and molecular analysis have been used to determine MGMT methylation status. This drives the demand for automated, non-invasive methods utilizing medical imaging.

This study proposes a deep learning-based approach to use multi-modal 3D brain MRI data to predict the methylation state of the MGMT promoter. Four MRI modalities—T1, T1-weighted contrast-enhanced (T1GD), T2, and FLAIR—are used by the model to provide complementary data regarding the tumor's anatomy and surrounding tissues. To efficiently process volumetric data and extract hierarchical spatial information, a new 3D DenseNet architecture was created.

Squeeze-and-Excitation (SE) attention techniques are integrated into the network to highlight pertinent feature channels in order to improve feature representation. To capture both broad and high-intensity features, a mixed pooling method that combines global average pooling and global max pooling is used. A gated fusion method is used to incorporate clinical characteristics like patient age and gender into the model in addition to imaging data, allowing the model to take use of multi-modal information.

To provide reliable evaluation, the model is trained using a stratified 5-fold cross-validation technique. To increase the performance and dependability of the model, a number of methods are used, including threshold tuning, temperature scaling probability calibration, and data augmentation. The suggested method yields a balanced macro F1-score of 0.6037, an accuracy of 63.36%, and an Out-of-Fold AUC of 0.6239.

Grad-CAM is used to depict the areas of MRI scans that affect the model's predictions in order to enhance interpretability. The findings show that the model concentrates on tumor locations that are clinically significant, giving its predictions more assurance.

Keywords: MGMT promoter methylation, glioblastoma (GBM), multi-modal MRI, 3D DenseNet, volumetric medical imaging, Squeeze-and-Excitation attention, clinical feature fusion, model calibration, Grad-CAM interpretability

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1. INTRODUCTION

1.1 Project Overview

One of the most aggressive and prevalent primary brain tumors in adults is glioblastoma. Rapid growth, high invasiveness, and a dismal prognosis are its defining characteristics, making an early and precise diagnosis crucial. Treatment planning is made more difficult by the intricacy of tumor anatomy and patient variability.

The MGMT (O6-methylguanine-DNA methyltransferase) methylation status is one of the important indicators used in the clinical evaluation of glioblastoma. This biomarker is essential for assessing how well chemotherapy works, especially when it comes to forecasting how patients will react to alkylating drugs like temozolomide. Compared to patients with unmethylated genes, those with methylated MGMT genes typically respond better to treatment.

Traditionally, invasive techniques like biopsy followed by molecular or genetic testing are used to evaluate the MGMT methylation status. Although these techniques are dependable, they are costly, time-consuming, and may put patients at danger. Consequently, there is an

increasing demand for effective and non-invasive alternatives that can help forecast this biomarker using medical imaging data.
It is now feasible to extract significant patterns from complicated imaging data, such MRI scans, thanks to the development of deep learning algorithms, especially in the field of medical image analysis. This project proposes a deep learning-based method to use multi-modal 3D brain MRI data to predict MGMT methylation status.

The suggested method extracts spatial features from volumetric input using a 3D DenseNet neural network architecture. A variety of MRI modalities, such as T1, T1GD, T2, and FLAIR, are employed to offer supplementary information regarding the features of surrounding tissue, contrast enhancement, and tumor structure. To further improve prediction performance, clinical characteristics like gender and age are added to the model.
The objective of this study is to provide a dependable and effective system that can help doctors make well-informed judgments by offering a non-invasive way to forecast MGMT methylation status. This strategy may enhance treatment planning and lead to better patient outcomes.

1.2 Reason for Choosing the Project

The growing significance of artificial intelligence in the fields of medical imaging and healthcare is the driving force behind the choice of this project. Brain tumors, especially glioblastoma, are among the most aggressive and complicated types of cancer; to increase patient survival rates, prompt diagnosis and treatment planning are essential.

The MGMT methylation status is one of the main variables affecting glioblastoma therapy choices. This biomarker is essential for predicting a patient's response to chemotherapy. However, invasive procedures like biopsy followed by genetic analysis are the basis of current methods for identifying MGMT status. These treatments can be dangerous, time-consuming, and not necessarily possible for every patient.
The potential to create non-invasive techniques for identifying significant clinical biomarkers is expanding due to the quick developments in deep learning, particularly in the analysis of medical pictures. Rich spatial information regarding the structure of tumors and adjacent

tissues can be found in MRI images, which are frequently employed in clinical practice. Deep learning models can be used to leverage this data and offer insightful information without requiring intrusive testing.

This project was selected to investigate the use of deep learning methods for multi-modal MRI data analysis, particularly a 3D DenseNet-based architecture. The model can gather complementary information about tumor features, like contrast enhancement and edema, thanks to the utilization of different MRI modalities.

Additionally, by fusing imaging data with patient-specific information, the model's incorporation of clinical characteristics like age and gender offers a chance to enhance prediction performance. This is consistent with current medical AI approaches, which integrate multi-modal data to improve model efficacy.

From an academic standpoint, this project offers practical experience managing volumetric data, creating deep learning architectures, and assessing model performance using relevant metrics. Additionally, it provides exposure to real-world issues like interpretability, model calibration, and class imbalance.

Ultimately, the project was chosen to use cutting-edge machine learning techniques to address a clinically important issue with the aim of advancing non-invasive and effective diagnostic support systems in healthcare.

1.3 Societal Impact

By offering a non-invasive and clever method for predicting MGMT methylation status in glioblastoma patients, the suggested technology has a substantial potential impact on the healthcare industry. Early and precise detection of this biomarker is essential for directing treatment choices, especially when assessing the efficacy of chemotherapy. The approach can help clinicians make more prompt and informed judgments by enabling such predictions using MRI data.

Reducing reliance on invasive procedures like biopsies is one of this approach's main advantages. Surgical intervention is required for biopsies, which entails risks like bleeding, infection, and patient pain. MRI-based analysis, on the other hand, is frequently carried out in clinical settings and is non-invasive. Therefore, by reducing the need for extra diagnostic procedures, the suggested approach may enhance patient safety and comfort.

Economically speaking, the system can help down healthcare expenses. Conventional molecular testing techniques can be costly and require specific laboratory infrastructure. The cost of diagnosis and analysis can be greatly decreased by using computer models and already-existing MRI data. This is especially helpful in settings with limited resources, when access to cutting-edge lab facilities could be restricted.

Healthcare workers may benefit from the use of artificial intelligence into medical imaging by having less work and being more productive. Automated algorithms can help radiologists who frequently handle massive amounts of imaging data by highlighting significant regions of interest and offering preliminary observations. This can lessen the possibility of human error and aid in a quicker diagnosis.

Additionally, these devices can make advanced diagnostic support more accessible in underprivileged or rural areas. AI-based solutions can be very helpful in assessing medical data and supporting therapeutic choices in places where skilled specialists are not easily accessible.

From a wider angle, this research advances healthcare solutions powered by AI. It illustrates the potential of integrating imaging and clinical data for better predictive performance and shows how deep learning techniques may be applied to challenging medical issues.

In general, this initiative will improve patient care, increase the effectiveness of diagnosis, lower expenses, and encourage the use of intelligent systems in healthcare.

1.4 Justification of Project Title

The project's title, "MGMT Methylation Prediction using 3D MRI and Deep Learning," appropriately captures the main goal, approach, and extent of the suggested system. Each element of the title emphasizes the interdisciplinary nature of the project and reflects a crucial part of the work.

The phrase "MGMT Methylation Prediction" highlights the project's main objective, which is to forecast the MGMT gene's methylation status. Due to its direct impact on treatment planning and patient prognosis, this biomarker has great clinical value in glioblastoma. The initiative tackles a pertinent and significant issue in the world of medical diagnostics by concentrating on MGMT methylation.

The usage of volumetric medical imaging data is indicated by the title's reference to "3D MRI." In contrast to conventional 2D methods, 3D MRI offers precise spatial information regarding the shape, structure, and surrounding tissue of tumors. This makes it possible to analyze tumor characteristics in greater detail. By collecting various characteristics of the tumor, such as contrast enhancement and edema, the utilization of multiple MRI modalities significantly improves the richness of input data.
"Deep Learning" refers to the application of sophisticated computational methods to automatically extract intricate patterns from the data. In this study, hierarchical features are extracted from volumetric MRI data using a 3D DenseNet-based architecture. Because deep learning models can handle high-dimensional data and learn non-linear correlations, they are especially well-suited for this purpose.
The research also includes clinical characteristics like gender and age, which enhance the system's predictive power. This integration shows the multi-modal character of the method and is consistent with current medical AI approaches, even though it isn't stated in the title.

All things considered, the project title successfully communicates the main elements of the system, such as the medical goal, the kind of data utilized, and the computational approach. It maintains relevance to both the technical and medical realms while offering a clear and succinct depiction of the work.

1.5 Objectives

This project's main goal is to create a clever, non-invasive system that uses deep learning methods and multi-modal 3D MRI data to predict MGMT methylation status. The following is a summary of the project's specific goals:

- To create and put into practice a deep learning model that can interpret volumetric MRI data and extract significant spatial characteristics using a 3D DenseNet architecture.
- To use multi-modal MRI inputs, such as T1, T1GD, T2, and FLAIR sequences, to obtain complementary data regarding surrounding tissue properties, tumor shape, and contrast enhancement.

- To enhance predictive performance by integrating clinical characteristics like age and gender into the model and fusing them with imaging information using an efficient fusion method.
- To ensure reliable performance evaluation and effective use of the supplied dataset, the model will be trained and assessed using stratified 5-fold cross-validation.
- To correct class imbalance and enhance model performance using suitable methods such threshold optimization and weighted loss functions.
- To utilize probability calibration techniques like temperature scaling to improve the accuracy of model predictions.
- To give a thorough evaluation of the model's classification capability by employing a variety of measures, such as AUC, accuracy, and F1-score.
- To use visualization tools like Grad-CAM, which emphasize the areas of the MRI scans influencing the model's predictions, to study model behavior and enhance interpretability.
- To examine the suggested approach's shortcomings and pinpoint possible enhancements for further research.

Overall, the objective of this project is to develop a reliable and interpretable system that can assist in clinical decision-making by providing insights into MGMT methylation status using non-invasive imaging data.

2. CURRENT STUDY AND LITERATURE REVIEW

2.1 Overview

The use of deep learning techniques in medical imaging has received a lot of interest lately, especially for problems including prognosis prediction, tumor identification, and categorization. Researchers are now able to create models that can extract intricate patterns

from medical data thanks to the availability of large-scale imaging datasets and improvements in computing power.
One of the most aggressive brain cancers, glioblastoma, has been thoroughly investigated using imaging-based methods. In particular, because of its clinical significance in treatment planning, the prediction of MGMT methylation status using MRI data has become a significant research problem.
The analysis of brain MRI scans using machine learning and deep learning methods has been investigated in a number of research. Conventional methods depended on manually created features and traditional machine learning algorithms like random forests and support vector machines. Nevertheless, these techniques frequently have trouble capturing the intricate spatial patterns found in volumetric data.
Researchers can now automatically construct hierarchical feature representations from raw MRI data because to the development of deep learning, particularly convolutional neural networks (CNNs). More modern methods make use of 3D CNN architectures, which are more appropriate for volumetric data since they maintain spatial links in all dimensions. Multi-modal learning, which combines several MRI sequences to enhance model performance, has also attracted increasing attention. To improve prediction accuracy, the combination of clinical characteristics with imaging data has also been investigated.

Notwithstanding these developments, a number of issues still exist, including as small datasets, class imbalance, and tumor feature variability. These difficulties show how important it is to have reliable and understandable models.
In order to lay the groundwork for the suggested study, this chapter examines current methods for MGMT methylation prediction and identifies their shortcomings.

2.2 Review of Existing Methods

The examination of brain tumors using medical imaging has been extensively studied, with a special emphasis on glioblastoma. Numerous methods have been put forth to predict tumor features, including MGMT methylation status, thanks to the growing availability of MRI data and developments in machine learning.

Traditional machine learning methods were the mainstay of early research in this field. Using these methods, handcrafted features like texture, intensity, and shape-based descriptors were extracted from MRI data. Tumor regions were frequently represented by characteristics such as wavelet-based features, gray-level co-occurrence matrices (GLCM), and histogram statistics. Classical classifiers like Support Vector Machines (SVM), Random Forests, and Logistic Regression models were then given these attributes.

These techniques had a number of drawbacks even if they offered some degree of interpretability. The quality of feature engineering, which necessitated domain knowledge and was frequently time-consuming, had a significant impact on how well these models performed. Furthermore, the intricate spatial correlations found in volumetric MRI data were not always captured by hand-crafted features.

Data-driven feature learning has become more popular as deep learning, especially convolutional neural networks (CNNs), has grown in popularity. Without the requirement for human feature extraction, CNNs may automatically learn hierarchical feature representations from unprocessed images. Early deep learning techniques analyzed individual MRI slices individually using 2D CNN structures. These models outperformed conventional techniques, but they were unable to accurately depict the three-dimensional structure of tumors. Researchers developed 3D CNN architectures, which work with volumetric data and maintain spatial relationships in all three dimensions, in order to overcome this constraint. In tasks like tumor segmentation and classification, these models have demonstrated enhanced performance. Because 3D CNNs can extract contextual information from nearby slices, they are especially well-suited for interpreting MRI data.

The utilization of multi-modal MRI data is another significant development in recent research. Complementary information on tumor features is provided by several MRI modalities. For instance, T2-weighted images record fluid content, FLAIR images highlight edema, T1 contrast-enhanced (T1GD) images indicate areas of blood-brain barrier disruption, and T1-weighted images offer structural information. Combining these modalities enhances prediction performance and enables models to learn richer representations.

Numerous research have investigated the integration of clinical characteristics such patient age, gender, and genetic information in addition to imaging data. These characteristics can enhance the predictive power of models and offer more context. Recent studies using multi-modal learning techniques that integrate imaging and clinical data have shown encouraging outcomes.

The prediction of MGMT methylation status is still a difficult subject despite recent developments. The scarcity of labeled medical datasets is one of the main obstacles. Small datasets lower deep learning models' capacity for generalization and raise the danger of overfitting. Additionally, biased predictions may result from a class imbalance between methylated and unmethylated cases.
The inability of deep learning models to be interpreted presents another difficulty. Even if these models function well, it is crucial to comprehend their decision-making process in medical applications. Explainability methods like Grad-CAM, which offer visual insights into model behavior, have become popular as a result.
All things considered, current techniques have shown the promise of deep learning and machine learning for MGMT methylation prediction. However, constraints like the lack of data, the interpretability of the model, and the heterogeneity of tumor characteristics continue to spur additional study in this field.

Predicting MGMT methylation status using MRI data and deep learning algorithms has been the topic of several studies in recent years. The Brain Tumor Radiogenomic Classification challenge, which sought to predict MGMT methylation status from multi-modal MRI data, is one of the most well-known initiatives in this field. In this challenge, a number of strategies were put forth, including 2D and 3D deep learning models.

Convolutional neural networks trained on MRI slices or volumes were used by several participants in these research to extract features related to tumors. While some methods created unique architectures specifically for medical imaging tasks, others used transfer learning by utilizing pre-trained models. Models utilizing 3D volumetric data were found to perform better than 2D techniques because they gathered more contextual information and maintained spatial continuity.

Because DenseNet-based topologies can provide effective gradient flow and feature reuse, they have also been investigated in a number of research. Each layer can receive inputs from all preceding layers thanks to the dense connection structure, which lowers the possibility of vanishing gradients and aids in the learning of richer feature representations. DenseNet is

especially well-suited for medical imaging tasks with little datasets because of these characteristics.

For tumor-related tasks, other sophisticated architectures like ResNet and U-Net variations have been employed in addition to DenseNet. U-Net's encoder-decoder structure has been modified for classification jobs even though its primary purpose is segmentation. Strong performance in a variety of medical imaging applications has also been shown by ResNet-based models, which are well-known for their residual connections.

The utilization of attention processes is another significant trend noted in recent studies. The model can concentrate on the most essential areas or characteristics in the input data with the use of attention modules. To enhance feature representation, CNN designs have included strategies like channel attention and spatial attention. By reducing unimportant background information, these techniques enable the model to highlight significant tumor locations.

To increase prediction accuracy, researchers have also looked into combining imaging data with genomic and clinical data. Feature concatenation and attention-based fusion are two multi-modal fusion methods that have demonstrated encouraging outcomes. These methods seek to improve model performance by utilizing complimentary information from various data sources.

Despite these approaches' advancements, the outcomes of various investigations range greatly because of variations in model topologies, preprocessing methodologies, and dataset sizes. MGMT methylation prediction is still a difficult subject, as seen by the numerous studies that indicate mediocre accuracy.

The significance of appropriate evaluation techniques is another important finding from the literature. Due to the small dataset size, cross-validation is frequently used, however it may still produce inconsistent results between folds. This emphasizes the necessity of thorough assessment and cautious interpretation of performance indicators.

All things considered, the corpus of current research indicates that deep learning-based methods may be able to predict MGMT methylation status from MRI data. However, issues

including interpretability, model generalization, and limited data availability continue to motivate more study in this field.

2.3 Limitations of Existing Methods

Despite considerable advancements in the use of deep learning and machine learning techniques for MGMT methylation prediction, there are still a number of drawbacks with existing methods.

The lack of labeled medical data is one of the main obstacles. Deep learning models struggle to generalize well since most research datasets are rather small. Overfitting, in which the model performs well on training data but inconsistently on unseen data, is frequently the result of this.

Class imbalance, in which one class has much more samples than the other, is another problem. The minority class, which is frequently clinically significant, may perform poorly as a result of models becoming biased in favor of the dominant class.

Additionally, a lot of current methods are not interpretable. Because deep learning models are frequently seen as "black boxes," it can be challenging to comprehend how predictions are made. In medical applications, where transparency and trust are crucial, this is a crucial problem.

Model performance may also be impacted by variations in MRI acquisition procedures among various institutions. Inconsistencies in the data can be introduced by variations in scanner settings, resolution, and preprocessing techniques, which makes it more difficult for models to generalize across datasets.

Another drawback is that some models neglect clinical information in favor of relying solely on imaging data. Because clinical characteristics like age and gender may offer important context, this may lower prediction accuracy.

Lastly, a lot of research reports performance using constrained evaluation techniques, which could not accurately represent performance in the actual world. Optimistic estimates may result from the use of cross-validation and the lack of independent test sets.

The strategy suggested in this study is motivated by these restrictions, which emphasize the necessity for reliable, comprehensible, and well-generalized models.

3. PROJECT DESIGN

3.1 Contributions

Using multi-modal 3D MRI data, the proposed project presents a deep learning-based method for predicting MGMT methylation status. The following is a summary of this work's main contributions:

- To efficiently process volumetric MRI data and extract spatial features important for tumor characterisation, a novel 3D DenseNet architecture is created.
- T1, T1GD, T2, and FLAIR sequences are examples of multi-modal MRI inputs that are used to offer further information on the surrounding tissue and tumor structure.
- By concentrating on the most informative channels, Squeeze-and-Excitation (SE) attention methods are included into the network to improve feature representation.
- To enhance feature extraction and representation, a mixed pooling method that combines global average pooling and global max pooling is employed.
- A gated fusion method is used to merge clinical characteristics like age and gender with imaging features, enabling the model to efficiently utilize both kinds of data.
- To guarantee reliable assessment and effective use of the dataset, a stratified 5-fold cross-validation technique is used.
- To increase the accuracy and comprehensibility of model predictions, probability calibration methods such temperature scaling and threshold optimization are used.
- To display model attention and confirm that the model concentrates on clinically relevant regions, Grad-CAM is employed as an explainability tool.

All things considered, the research offers a thorough pipeline that tackles a difficult classification issue by fusing cutting-edge deep learning methods with clinical data and medical imaging.

3.2 Block Diagram of the system

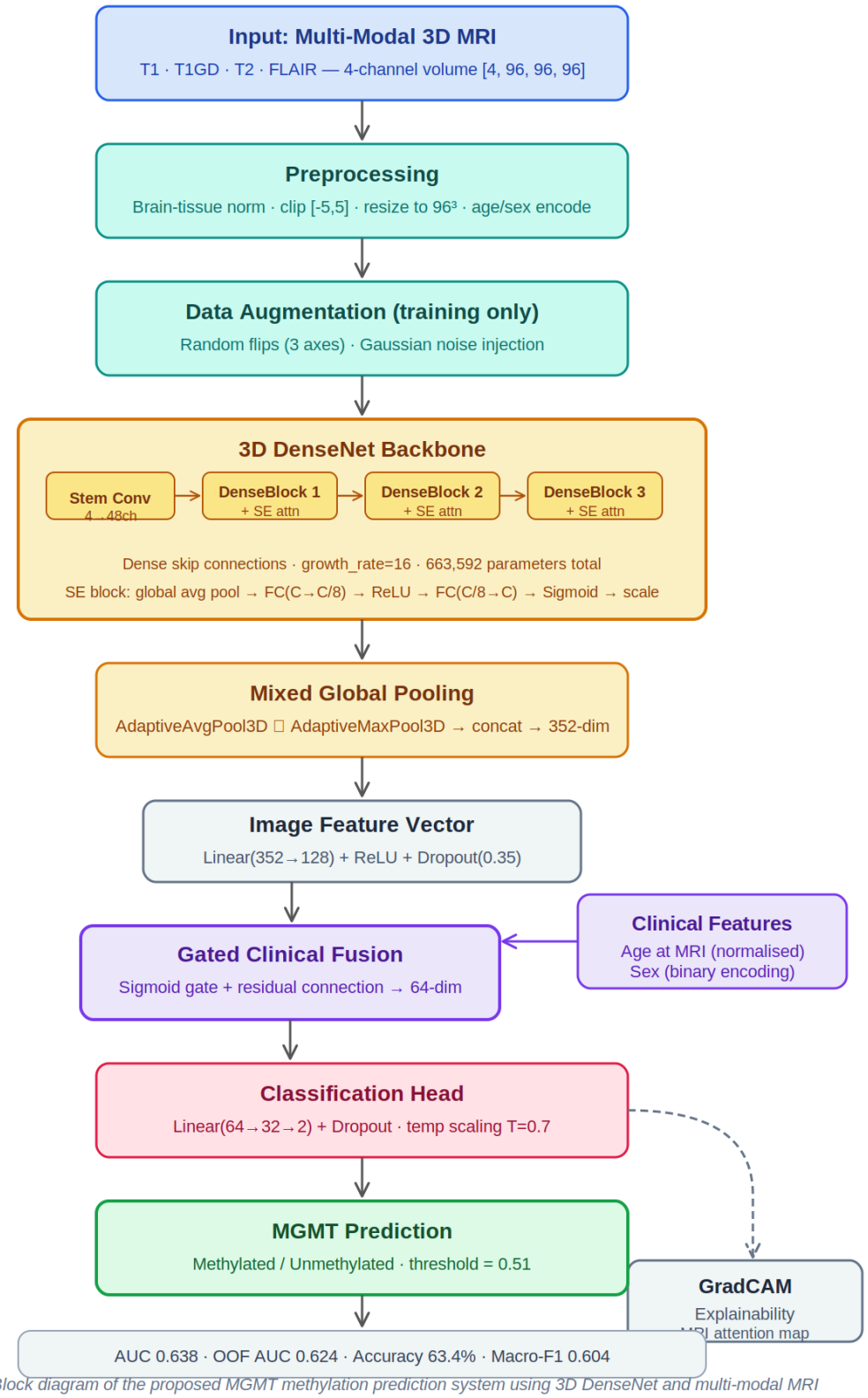

Figure 3.1: Block diagram of the proposed MGMT methylation prediction system using 3D DenseNet and multi-modal MRI

Figure 3.1: Block diagram of the proposed MGMT methylation prediction system using multi-modal 3D MRI data and a 3D DenseNet-based architecture.

3.3 Stages of Project Work

3.3.1 Data Acquisition and Understanding

The suggested MGMT methylation prediction methodology was developed in several clearly defined phases, guaranteeing a methodical and organized approach. From data preparation to final review and interpretation, every step is vital to the construction of the entire pipeline.

Acquiring and comprehending data is the first step. The UPenn-GBM dataset, which includes multi-modal MRI scans and related clinical data like age and gender, was utilized. Multiple MRI modalities, such as T1, T1GD, T2, and FLAIR, are included in every patient record. These modalities provide supplementary information regarding the tumor's anatomy and surrounding tissues.

3.3.2 Data Preprocessing

Preprocessing of the data is the main emphasis of the second stage. Normalization techniques were used to standardize the voxel intensity values because noise and intensity variations are common in medical imaging data. To eliminate background regions and concentrate on important tissue information, a brain-tissue-based normalization technique was employed. To guarantee uniformity among samples and compliance with the deep learning model, all MRI volumes were also downsized to a fixed dimension of $96 \times 96 \times 96$.

3.3.3 Data Augmentation

Data augmentation is the third step. In order to increase model generalization during training, augmentation approaches were used due to the dataset's small size. These methods include adding tiny Gaussian noise and randomly flipping along several spatial axes. Augmentation helps the model become more robust to variations in input data and reduces the risk of overfitting.

3.3.4 Model Design and Feature Extraction

Model design and feature extraction constitute the fourth phase. Volumetric MRI data was processed using a specially designed 3D DenseNet architecture. The network is made up of several Dense Blocks with skip connections that encourage effective gradient flow and feature reuse. In order to improve channel-wise feature representation and enable the model to concentrate on the most pertinent features, squeeze-and-excitation attention modules were incorporated.

3.3.5 Feature Aggregation and Fusion

Feature fusion and aggregation are part of the fifth stage. A mixed pooling method that combined global average pooling and global max pooling was used to capture both average and high-intensity feature responses after imaging features were extracted using the DenseNet backbone. A tiny neural network was used in parallel to process clinical parameters like gender and age. The model was then able to determine the relative significance of each modality by combining these features with imaging features through a gated fusion process.

3.3.6 Model Training

Model training is the main focus of the sixth stage. In order to ensure that each fold retains a balanced distribution of classes, the model was trained using a stratified 5-fold cross-validation technique. Robust examination and effective use of the provided dataset are made possible by this method. The training procedure was guided by suitable optimization techniques and loss functions.

3.3.7 Model Evaluation

Performance analysis and model evaluation are part of the seventh step. Several performance criteria, such as accuracy, F1-score, and Area Under the ROC Curve (AUC), were used to assess the model. Overall model performance throughout all validation folds was evaluated using out-of-fold (OOF) predictions.

3.3.8 Probability Calibration and Threshold Optimization

Threshold optimization and probability calibration are included in the eighth stage. In order to enhance the calibration of anticipated probability and guarantee more dependable confidence scores, temperature scaling was used. OOF predictions were used to estimate the ideal decision threshold in order to achieve a fair trade-off between recall and precision.

3.3.9 Model Interpretability

The last phase concentrates on the display and interpretability of the model. The areas of the MRI scans that contribute most to the model's predictions were highlighted in attention maps created using Grad-CAM, an explainability technique. This aids in confirming that the model is concentrating on areas that are clinically pertinent.

In order to solve the issue of MGMT methylation prediction, the project generally employs an organized pipeline that combines data preprocessing, deep learning-based feature extraction, multi-modal fusion, and interpretability methodologies

3.4 Tools and Technologies Used

A number of software tools, programming libraries, and hardware resources are needed to build the implemented MGMT methylation prediction system. These tools were chosen to handle medical imaging data effectively, create deep learning models, and carry out evaluation and visualization.

Python Programming Language

Because of its ease of use, adaptability, and broad support for machine learning and data analysis frameworks, Python was chosen as the main programming language for the entire project. The various components needed for data processing, model building, and evaluation may be seamlessly integrated thanks to its rich ecosystem.

PyTorch Deep Learning Framework

The model was implemented using PyTorch as the primary deep learning framework. It offers effective GPU acceleration, flexible model design, and dynamic computation graphs. Custom architectures, like the 3D DenseNet utilized in the project, can be easily implemented with PyTorch.

NumPy and Pandas

Data management and preprocessing were done using NumPy and Pandas. Pandas was used to handle tabular data, such as patient metadata like age, gender, and labels, while NumPy offers effective numerical operations on multi-dimensional arrays.

NiBabel Library

Medical imaging data in NIfTI (.nii) format was loaded and processed using NiBabel. It makes volumetric MRI data accessible and makes it easier to convert it into numerical arrays that the deep learning model can use as input.

Scikit-learn

Data splitting methods and evaluation criteria were implemented using Scikit-learn. Model performance was evaluated using functions like stratified cross-validation and performance indicators including AUC, accuracy, and F1-score.

Matplotlib

ROC curves, probability distributions, calibration plots, and confusion matrices were among the results that were visualized using Matplotlib. It facilitates the clear and understandable presentation of results.

CUDA and GPU Acceleration

The deep learning model was trained more quickly thanks to GPU acceleration. When dealing with complicated neural network topologies and big volumetric data, CUDA-enabled GPUs dramatically shorten calculation times.

Development Environment

Kaggle Notebook, which enables interactive code execution and simple output display, was used to implement the project. This environment is very helpful for iterative development, troubleshooting, and experimenting.

Operating System

The implementation was done in a cloud-based environment (Kaggle), which offers pre-installed libraries and GPU resources needed for data processing and deep learning activities. Windows operating system was used on the machine

Overall, the combination of these tools and technologies enabled efficient implementation, training, and evaluation of the proposed system.

4. IMPLEMENTATION

4.1 Overview

The suggested system's implementation focuses on employing a deep learning-based method to predict the MGMT methylation status from 3D brain MRI data. The algorithm incorporates pertinent clinical data and efficiently extracts spatial elements from volumetric medical images.

The foundation for feature extraction in this study is a 3D DenseNet architecture that has been improved with Squeeze-and-Excitation (SE) attention methods. The model develops hierarchical representations of tumor features by processing multi-modal MRI data. To enhance predictive performance, a fusion method is often used to incorporate clinical characteristics like gender and age into the network. .

Data preprocessing, stratified cross-validation model training, and evaluation using suitable performance metrics like AUC, accuracy, and F1-score comprise the full workflow. Using methods like temperature scaling and threshold optimization, special attention is paid to addressing class imbalance and enhancing probability calibration.

The PyTorch deep learning framework is used in the implementation, allowing for flexible custom model component design and effective training on GPU hardware.

4.2 Dataset Description

The data utilized in this research comes from the UPenn-GBM dataset, which includes multi-modal brain MRI scans from patients diagnosed with glioblastoma. Each patient entry encompasses volumetric MRI data accompanied by labels indicating the MGMT methylation status, which shows whether the tumor is methylated or unmethylated.

Several MRI modalities, including T1-weighted, T1-contrast enhanced (T1CE), T2-weighted, and FLAIR sequences, are included in the collection. In order to make an accurate prognosis, several modalities offer complementary information on the surrounding tissues, edema, and tumor shape.

This approach makes use of 262 patient samples in total. Every sample has a binary label, with class 0 denoting unmethylated MGMT status and class 1 denoting methylated MGMT status. With roughly 151 unmethylated cases and 111 methylated cases, the sample shows a modest class imbalance.

Basic clinical characteristics like patient age and gender are given in addition to imaging data. By adding context beyond imaging data, these traits are included into the model to improve predictive performance.

Each patient has a specific folder with preprocessed MRI volumes in the structured format of the dataset. Patient IDs are mapped to their matching labels and clinical characteristics using a metadata file. Effective data loading and integration during training and evaluation are made possible by this organized framework.

4.3 Data Preprocessing

As raw medical imaging data frequently contains intensity changes, noise, and uneven spatial dimensions, data preparation is essential to this job. To guarantee consistency and enhance model performance, an organized preprocessing pipeline is used.

Each patient's MRI data includes four modalities: T1, T2, FLAIR, and T1-weighted contrast-enhanced (T1GD). After being processed separately, these modalities are integrated to create a multi-channel input for the model.

Z-score normalization is applied exclusively to the foreground region (non-zero voxels) of each MRI volume. This maintains significant anatomical structures while ensuring that the intensity values are consistent across patients. To eliminate severe outliers and enhance training stability, the volumes are clipped to a predefined range of [-5, 5] after normalization.

All volumes are reduced to a standardized shape of 96 × 96 × 96 because MRI images may have different spatial dimensions. A center crop or padding technique is used to do this, guaranteeing that crucial areas are kept intact while preserving a constant input size throughout the dataset.
The deep learning model receives a 4-channel volumetric input of shape (4, 96, 96, 96), which is created by stacking the preprocessed modalities for each patient.
Clinical metadata is processed in addition to imaging data. Only baseline scans are included in the dataset, and only accurate MGMT labels—both methylation and unmethylated—are kept. Methylated is expressed as 1 and unmethylated as 0 in the binary label that represents the MGMT status. Age numbers are used as numerical inputs, and gender is encoded as a binary characteristic.
To maintain data consistency, patients with missing MRI modalities are eliminated. A metadata file that associates patient IDs with the appropriate imaging folders and labels is used to arrange the final dataset.
Additional checks, such as validation of volume shapes, normalization statistics, and visual inspection of sample slices, are carried out to confirm preprocessing correctness. This guarantees that the processed data is reliable and appropriate for model training.

4.4 Model Architecture

The proposed model is a 3D DenseNet architecture that was specifically created to handle volumetric brain MRI data and incorporate clinical data for MGMT methylation prediction.

Four MRI modalities (T1, T1GD, T2, and FLAIR) are combined to create a 4-channel volumetric input with dimensions of 4 × 96 × 96 × 96. The model can capture complementary information regarding surrounding edema, contrast enhancement, and tumor anatomy thanks to this multi-modal input.

The design starts with a convolutional stem layer that uses a 3D convolution to increase the input channels from 4 to 48. Batch normalization, ReLU activation, and max pooling come next. This first step aids in lowering spatial resolution and extracting low-level spatial characteristics.

Several Dense Blocks, each of which has several densely connected convolutional layers, make up the network's core. This topology enhances gradient flow and feature reuse since every layer in the block receives input from every layer before it. The model can learn rich hierarchical representations while retaining parameter efficiency because to this approach.

Squeeze-and-Excitation (SE) attention modules are added after each Dense Block to improve feature representation. By determining the significance of each feature map and recalibrating them appropriately, these modules execute channel-wise attention. This aids the model in concentrating more on pertinent traits associated with tumors.

Transition layers are used in between Dense Blocks to downsample the spatial dimensions and decrease the number of feature channels. Batch normalization, ReLU activation, a 1x1x1 convolution, and average pooling make up these layers.

Global average pooling and global max pooling are used to implement a mixed pooling approach following the final Dense Block. A richer feature representation is produced by concatenating the outputs of various pooling processes to incorporate both strong activation signals and average feature responses.

To create a compact feature vector, the recovered imaging features are subsequently run through a fully connected layer. A tiny multi-layer perceptron (MLP) is used in parallel to process clinical characteristics like gender and age, converting them into a learnt feature representation.

Imaging and clinical characteristics are combined using a gated fusion method. In order to balance the contributions of imaging and clinical data, this module develops adaptive weights. To maintain the original combination features and enhance performance and stability, a residual connection is also added.

Lastly, a classification head with dropout regularization and fully connected layers processes the fused feature vector. Two logits representing the methylated and unmethylated classes are generated by the output layer.

Overall, the architecture improves prediction performance for MGMT methylation status by combining multi-modal imaging signals with clinical data, strengthened by attention mechanisms and adaptive fusion.

4.5 Training Strategy

To guarantee reliable assessment and effective use of the supplied dataset, the model is trained using a stratified 5-fold cross-validation technique. This method divides the dataset into five folds while preserving the distribution of classes in each split. One fold is utilized for validation and four folds are used for training in each iteration.

PyTorch DataLoader objects are used to load data, and only the training set is subjected to data augmentation. The augmentation, which enhances model generalization, consists of random flipping along various spatial axes and minor intensity fluctuations. .

The AdamW optimizer, which combines adaptive learning rates with weight decay for improved generalization, is used to optimize the model. During training, the learning rate is gradually lowered using a cosine annealing-based learning rate scheduler to facilitate smoother convergence.

Class weights are dynamically calculated based on the distribution of labels in the training fold to correct the dataset's class imbalance. In order to guarantee that both classes contribute equally during training, these weights are integrated into the cross-entropy loss function.

To avoid overfitting, early halting is used during the training phase, which lasts for a predetermined number of epochs. Validation AUC is used to track the model's performance, and anytime an improvement in validation AUC is noted, the optimal model checkpoint is kept.

In order to stabilize training and stop gradients from bursting, gradient clipping is used during backpropagation. This guarantees that model parameters are updated consistently.

To enhance probability calibration, temperature scaling is used to the model outputs during validation and inference. To improve class separation and sharpen the probability distribution without interfering with training, a temperature value of $T = 0.7$ is employed.

The best-performing model is utilized to produce predictions on the validation set at the conclusion of each fold. Out-of-fold (OOF) performance metrics, which offer a trustworthy assessment of the model's capacity for generalization, are calculated by adding these predictions across all folds.

4.6 Loss Function and Optimization

The cross-entropy loss function, which is appropriate for binary classification problems like MGMT methylation prediction, is used to train the model. Class weights are added to the loss function to correct the dataset's class imbalance and guarantee that both classes contribute equally during training.

The AdamW optimizer is used to optimize the model's parameters. This optimizer improves generalization and avoids overfitting by combining adaptive learning rates with weight decay regularization.

During training, the learning rate is progressively lowered using a cosine annealing learning rate scheduler. This enhances stability in later training phases and enables the model to converge more smoothly.

In order to avoid exploding gradients and provide steady updates to the model parameters, gradient clipping is used during backpropagation. In order to lessen overfitting and enhance generalization performance, the network additionally has dropout layers.

Furthermore, the estimated probabilities are calibrated during inference by applying temperature scaling. This enhances the model's output reliability without interfering with training.

4.7 Evaluation Methodology

To ensure a thorough evaluation of the proposed model's classification capabilities, its performance is assessed using a variety of indicators. Both threshold-independent and threshold-dependent metrics are taken into consideration because MGMT methylation prediction is a binary classification problem.

The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is the main evaluation metric. This score assesses how well the model can differentiate between classes that are methylated and those that are not across a range of threshold values. Better separability between the two classes is indicated by a higher AUC.

The overall percentage of correctly identified samples is measured using accuracy in addition to AUC. However, accuracy alone is insufficient to assess model performance due to the dataset's small imbalance.

The F1-score, which is the harmonic mean of precision and recall, is used to address this. To guarantee that both classes are given equal weight, the macro F1-score is taken into account. Classification performance can also be visualized using confusion matrices. They aid in understanding model bias towards any given class by offering insights into true positives, true negatives, false positives, and false negatives.

A decision threshold is necessary to translate the model's output probabilities into class labels. Out-of-fold (OOF) predictions are used to determine an ideal threshold rather than a default threshold of 0.5. In order to maximize the balanced macro F1-score and guarantee that both classes have enough recall, this level is used.

Temperature scaling is applied to the model outputs during inference to improve probability calibration. This guarantees that each class's actual likelihood is more accurately reflected in the forecasted probabilities.

Out-of-fold predictions from the cross-validation procedure are used to calculate all assessment measures. This gives a trustworthy assessment of how well the model generalizes to new data.

4.8 Explainability using Grad-CAM

In order to improve the interpretation of the proposed model, gradient-weighted class activation mapping (Grad-CAM) is used as a post-hoc explanation technique. Grad-CAM provides visual insight into the regions of the input MRI volume that contribute most to the prediction of the model.

In this implementation, a 3D Grad-CAM approach is used, extracting gradients from the network's final density blocks. In particular, the Squeeze-and-Excitation (SE) layer in the last dense block is selected as the target layer, as it contains high semantic characteristics relevant for classification.

During inference, the predicted class score gradient is calculated relative to the feature map. These gradients are global averaged to obtain important weights that are then used to calculate a weighted combination of feature maps. The resulting activation map highlights the regions that have the greatest influence on the decision-making process of the model.

Since the Grad-CAM output has a lower spatial resolution, it is increased by using trilinear interpolation to correspond to the original input size (96 96 96). The generated 3D attention maps are then visualized by overlaying them on selected MRI slices, allowing a clear interpretation of the focus areas of the model.

Grad-CAM analysis is performed with the best folding model saved during training without requiring any retraining. Representative samples from both methylated and unmethylated classes are selected to analyze the behavior of the model.

This approach enables the qualitative validation of the model by verifying whether it focuses on clinically relevant areas such as tumor cores and surrounding edema. Grad-CAM is only used to draw conclusions, and does not affect the training process and serves as a tool to improve the trust and interpretation of model predictions.

5. RESULTS AND ANALYSIS

5.1 Overview

This chapter presents the experimental results obtained from the proposed model for the prediction of MGMT methylation. The model's performance is evaluated using multiple parameters, including AUC, accuracy, and F1 score, to provide a comprehensive understanding of its efficiency.

The results are obtained by using a stratified five-fold cross-validation strategy to ensure that the evaluation is robust and impartial. Out-of-fold (OOF) prediction is used to calculate overall performance indicators and provides reliable estimates of the generalization capabilities of the model.

In addition to quantitative evaluation, visual analysis techniques such as confusion matrices, ROC curves and GradCAM visualizations are used to interpret model behavior. These analyses help to understand the strengths and limitations of the proposed approach.

The following sections show detailed results and analyses based on the obtained experimental results.

5.2 Quantitative Results

Using the UPenn-GBM data set and a five-fold stratified cross-validation approach, the suggested DenseNet-based 3D model's performance was assessed. Because of the magnitude of the dataset, the results demonstrate consistent learning behaviors in all folds with modest variability.

The corresponding AUC values for each quarter were 0.5513, 0.5275, 0.6182, 0.7485, and 0.7455. While Fold 1 and Fold 2 showed very poor performance, the model performed best in Fold 4 and Fold 5, both surpassing an AUC of 0.74. This shift suggests that some data set subsets are consequently more challenging to categorize.

Overall, the model's accuracy was 0.6264 0.0498 and its cross-validation AUC was 0.6382 0.0937. The model's generalization performance on invisible data is reliably estimated by the AUC outside the fold (OOF), which is 0.6239 when all validation layers' predictions are combined.

In terms of classification performance, the model obtained a macro F1 balanced score of 0.6037 at the optimal decision threshold of 0.51 and an accuracy of 63.36%. These findings suggest that the model has a moderate ability to differentiate between the methyl and non-methyl classes. Squeeze-and-Excitation-Attention Mechanisms, multi-modal MRI inputs, and the integration of clinical characteristics are all responsible for performance gains. Additionally, the use of temperature scales and the removal of label smoothness improved class separation and probability calibration.

Fold	**AUC**	**Accuracy**
Fold 1	0.5513	0.5660
Fold 2	0.5275	0.5660
Fold 3	0.6182	0.6538
Fold 4	0.7485	0.6731

Fold 5	0.7455	0.6731
Mean +- std	0.6382 +-0.0937	0.6264 +-0.0498

Table 5.1 AUC and accuracies in different folds

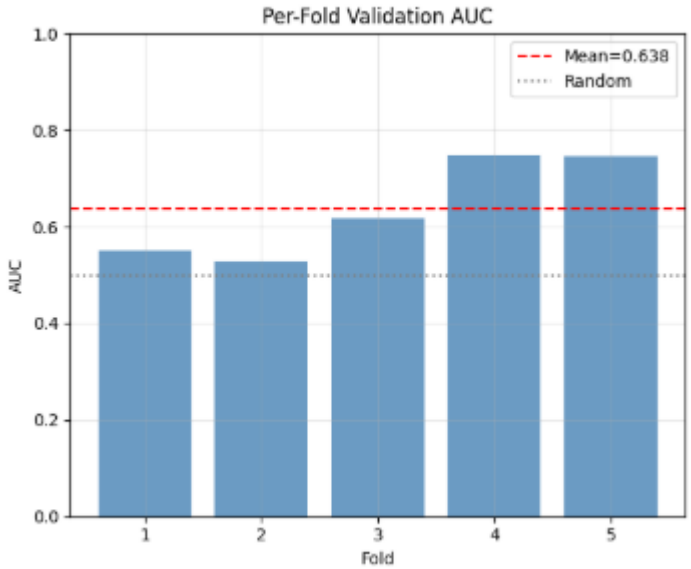

Fig 5.1: Bar chart of AUC at different folds

The variation in performance across folds highlights the challenge of MGMT methylation prediction from MRI data, where certain patient subsets exhibit weaker or less distinguishable imaging patterns, leading to lower classification performance in specific folds.

5.3 Confusion Matrix Analysis

The model's classification performance at the optimal decision threshold of 0.51 is shown in detail by the confusion matrix. According to the findings, the model is more effective at recognizing unmethylated cases than methylated ones.

The model's recall for the unmethylated class was roughly 79%, meaning that the majority of unmethylated samples were properly identified. The recall for the methylated class, on the other hand, was almost 42%, indicating that a sizable portion of methylated cases were incorrectly classified.

This recall imbalance suggests that the model has a small bias in favor of the majority class (unmethylated). However, threshold optimization prevents total domination of one class by ensuring that both classes maintain a respectable level of recall.

False negatives (methylated anticipated as unmethylated) are more common than false positives, according to the confusion matrix. This emphasizes the intrinsic challenge of identifying methylated instances, which might not show distinctive characteristics in MRI images.

Overall, the confusion matrix shows that even while the model performs in a balanced manner, more work needs to be done to increase sensitivity to the methylation class.

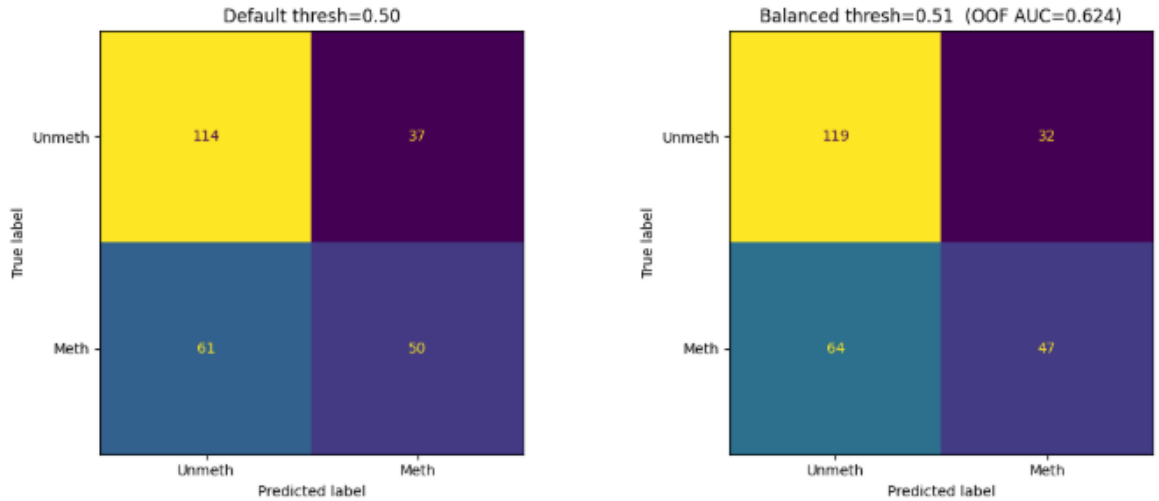

Figure 5.2: Confusion matrix both at default and balanced threshold

5.3.1 Threshold Comparison Analysis

The default threshold (0.50) and the optimized threshold (0.51) derived from out-of-fold predictions were compared in order to assess the effect of decision threshold selection on model performance.

The model produced an accuracy of 63% at the default threshold of 0.50, with a recall of 45% for the methylated class and 75% for the unmethylated class. This suggests that while a significant portion of methylation cases are incorrectly identified, the model does a better job of recognizing unmethylated cases.

There is a minor improvement in class balance following the application of the optimized threshold of 0.51. While the recall for the methylation class somewhat declines to 42%, the

recall for the unmethylated class rises to 79%. At 63%, the total accuracy is essentially unchanged.
At the optimal threshold, the macro F1-score slightly improves, suggesting a better balance between recall and precision in both classes. This shows that the trade-off between sensitivity and specificity can be affected by even minor changes in threshold selection.
In general, the optimized threshold offers a better balanced classification performance, especially when it comes to preserving a respectable recall for both classes, which is crucial for tasks involving medical decision-making.

Metric	**Default(0.50)**	**Optimised(0.51)**
Accuracy	0.63	0.63
Unmeth recall	0.75	0.79
Meth recall	0.45	0.42
Macro F1	0.60	0.60

Table 5.2 Metric comparison and thresholds

5.4 ROC Curve Analysis

The model's capacity to discriminate between methylation and unmethylated classes across various decision thresholds is assessed using the Receiver Operating Characteristic (ROC) curve. It is a trustworthy evaluation metric for classification tasks since it offers a threshold-independent performance measure.
The mean ROC curve and the ROC curves for each fold are displayed in Figure X. The performance of each fold curve varies, with Fold 4 and Fold 5 obtaining noticeably higher AUC values than Fold 1 and Fold 2. This indicates that the model performs better on certain subsets of the dataset, while some subsets are inherently more difficult to classify.

An overview of the model's performance is given by the mean ROC curve, which is colored to show the standard deviation. The model outperforms random classification (AUC = 0.5) with a mean AUC of 0.6382, indicating a moderate level of discriminative ability.

The out-of-fold (OOF) ROC curve, which is created using predictions from every validation fold, is also examined. An accurate assessment of the model's performance on unknown data is given by the OOF AUC of 0.6239.

The ROC curve also highlights the operational point that was chosen through threshold tuning. The model obtains a true positive rate (TPR) of roughly 0.423 and a false positive rate (FPR) of roughly 0.212 at the selected threshold of 0.51. This suggests a trade-off between specificity and sensitivity.

Although there is potential for development in reaching higher separability, the ROC analysis generally reveals that the model can distinguish between classes more accurately than random guessing.

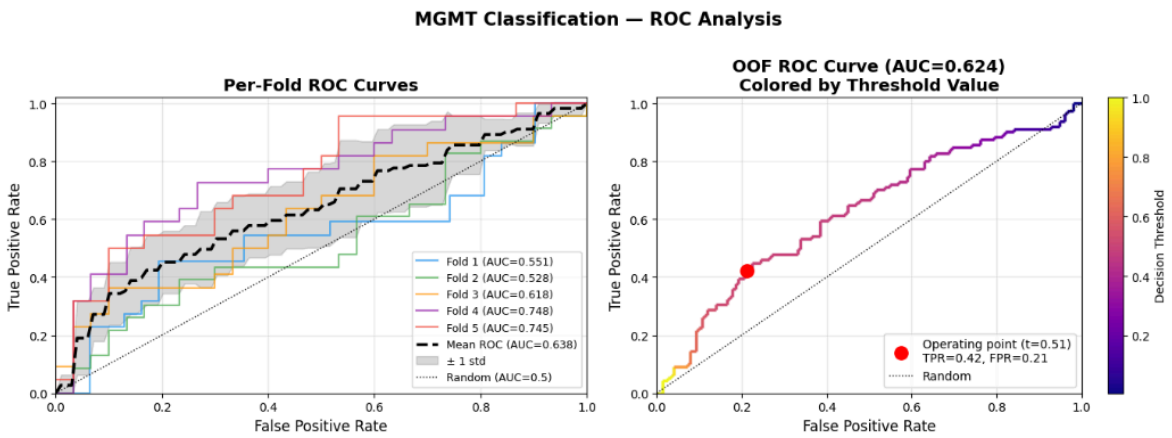

Figure 5.3: ROC curves for all folds and OOF ROC curve showing model performance across different thresholds.

5.5 Probability Distribution and Calibration Analysis

The distribution of projected probabilities for both classes is examined in order to gain a better understanding of the model's prediction behavior. The model's degree of confidence in differentiating between methylation and unmethylated situations is revealed by the probability distribution.

With a mean of 0.4576 and a standard deviation of 0.2151, the overall probability distribution suggests a rather wide range of forecasts. The values show that the model can generate both low-confidence and high-confidence predictions, ranging from close to 0 (0.0060) to 1.0000.

The projected odds for unmethylated cases have a mean of 0.4274 when examined independently, whereas the mean for methylated cases is marginally higher at 0.4986. Although the overlap reveals that the task is still difficult, the gap between the two class distributions ($\Delta\mu = 0.0712$) shows that the model can distinguish between classes.

The mediocre classification performance can be explained by the probability histograms, which demonstrate a substantial overlap between the two classes. Misclassifications result from this overlap, especially in borderline situations where estimated probabilities are near the decision threshold.

The Expected Calibration Error (ECE) is used in calibration analysis to assess the accuracy of anticipated probability. Although there is still considerable divergence, the calculated ECE value of 0.1153 shows moderate calibration, which means that the predicted probabilities fairly represent the true chances of each class.

By modifying the prediction confidence, the use of temperature scaling during inference aids in calibration. Better probability distribution and more dependable findings follow from this.

Overall, the probability distribution and calibration analysis show that the model generates significant confidence scores, but they also draw attention to how challenging the classification problem is because of overlapping feature distributions.

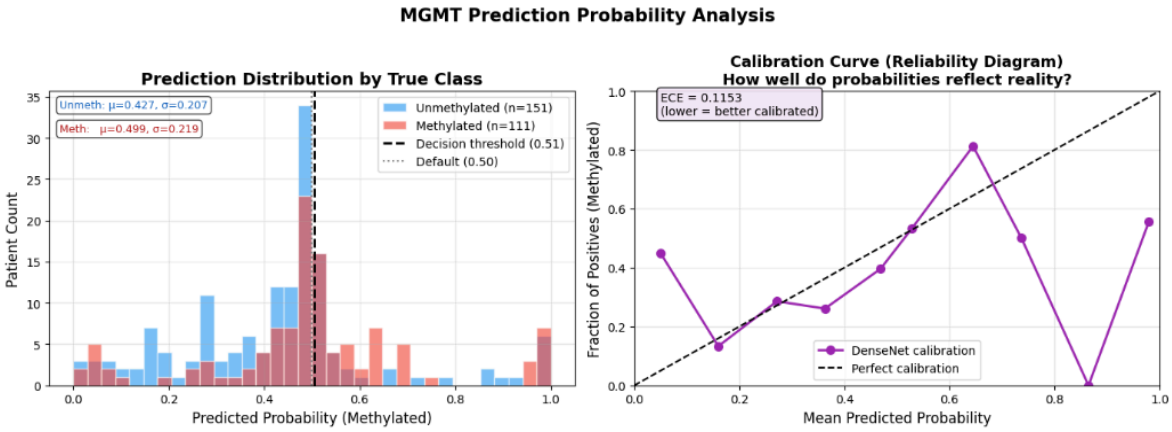

Fig 5.4. Prediction Probability Analysis

5.6 GradCAM Analysis

Grad-CAM visualization is used to highlight the areas of the MRI scans that contribute more to the predictions in order to interpret the decision-making process of the suggested model.

The Grad-CAM results demonstrate that the model concentrates on particular brain regions that are important for classifying MGMT methylation. The model identifies certain areas that match the tumor's structural characteristics in unmethylated samples.
In several instances, the activation maps for methylated samples are relatively less intense, suggesting a lower level of prediction confidence. The model has comparatively reduced recall for the methylated class, which is consistent with the reported performance measures.

Additionally, the visualizations show that the model focuses on locations that are clinically significant for glioblastoma analysis, such as those close to the tumor core. This implies that rather than depending on unrelated background data, the model is learning significant features.
Nevertheless, in other instances, the Grad-CAM maps display mild or diffuse activations, suggesting prediction uncertainty. This emphasizes even more how challenging the endeavor is and how more data or better feature extraction techniques are required.
Overall, Grad-CAM increases confidence in the model's predictions by qualitatively validating the model by showing that it concentrates on medically relevant regions.

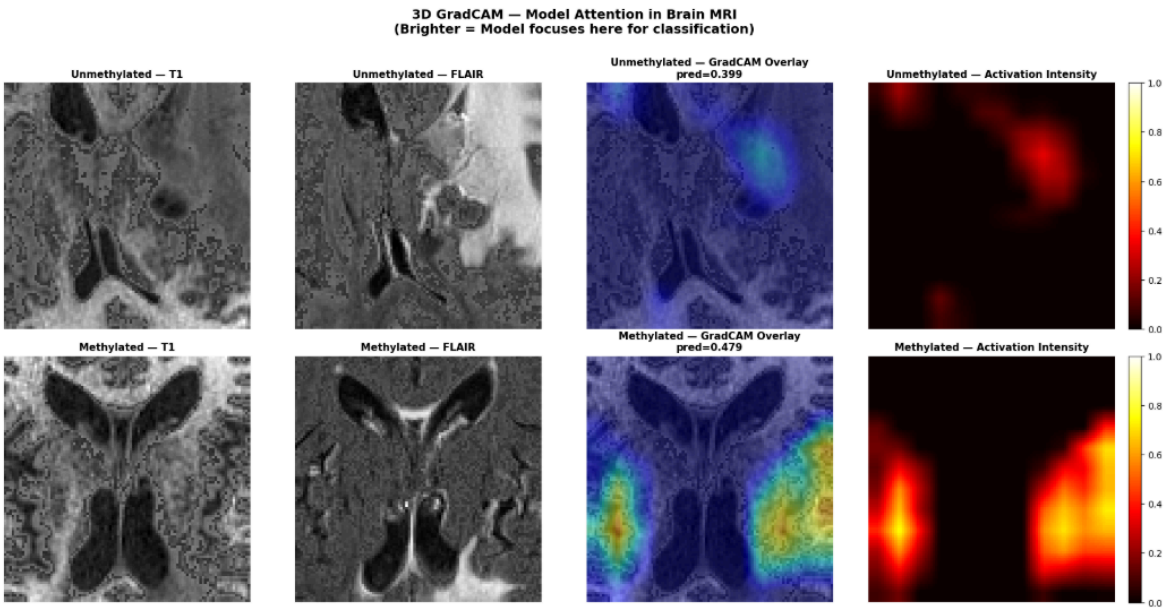

Figure 5.5: GradCAM visualization showing regions of interest for methylated and unmethylated cases.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This study used multi-modal 3D brain MRI data to construct a deep learning-based method for predicting MGMT methylation status. Imaging features and clinical data were combined using a proprietary 3D DenseNet architecture that was improved with Squeeze-and-Excitation attention and a gated fusion technique.

Using the UPenn-GBM dataset, stratified 5-fold cross-validation was used to assess the model. With an out-of-fold AUC of 0.6239 and a mean AUC of 0.6382 ± 0.0937, it demonstrated considerable discriminative power. The application of temperature scaling and threshold optimization enhanced probability calibration and resulted in a more balanced classification performance.

The findings demonstrate that although the model is able to extract significant patterns from MRI data, the task is still difficult because of the small sample size and overlapping feature distributions between classes. The validity of learned features was further supported by Grad-CAM analysis, which showed that the model concentrates on clinically significant regions like the tumor core and adjacent areas.
Overall, the suggested method shows the value of integrating multi-modal imaging with clinical data and offers a respectable baseline for MGMT methylation prediction.

6.2 Future Scope

Future work could improve the suggested system in a number of ways. Enhancing generalization and lowering variability among folds can be achieved by expanding the dataset or adding new datasets.
To capture long-range spatial dependencies in volumetric data, more sophisticated architectures like transformer-based models or hybrid CNN-transformer techniques can be investigated. Prediction performance may be further improved by adding more clinical or radiomic features.
To increase robustness, test-time augmentation (TTA) and data augmentation approaches can be used. Furthermore, forecasts made using ensemble approaches that include several models may be more reliable and accurate.
More sophisticated explainability strategies can be investigated from an interpretability standpoint to offer more profound understandings of model behavior. Lastly, to assess the model's practicality, incorporating it into a clinical decision support system can be taken into consideration.

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Appendix A

A.1 Dataset Directory Structure

The dataset used in this project follows a structured directory format, where each patient folder contains multiple MRI modalities.

```
processed_upenn/
├── UPENN-GBM-00001/
│   ├── T1.nii
│   ├── T1GD.nii
│   ├── T2.nii
│   ├── FLAIR.nii
├── metadata_upenn.csv
```

Each patient folder contains four MRI modalities, which are used as input channels for the model.

A.2 Dataset Loading and Preprocessing

The following code snippet shows the dataset class used for loading MRI volumes and clinical features.

```
class MGMTDataset(Dataset):
    def __init__(self, dataframe, augment=False):
        self.df = dataframe.reset_index(drop=True)
        self.augment = augment

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        modalities = []
        for mod in ['T1', 'T1GD', 'T2', 'FLAIR']:
            path = os.path.join(row['folder_path'], f'{mod}.nii')
            img  = nib.load(path).get_fdata(dtype=np.float32)
```

```
            # Brain-tissue-only normalization (excludes background
voxels)
            mask = img > 0
            if mask.sum() > 100:
                mean = img[mask].mean()
                std  = img[mask].std() + 1e-8
                img  = (img - mean) / std
                img[~mask] = 0.0
            else:
                img = (img - img.mean()) / (img.std() + 1e-8)

            # ✅ NEW: Clip extreme values — removes rare outlier voxels
            # after normalization some voxels reach ±10–15 std,
destabilizing training
            img = np.clip(img, -5.0, 5.0)

            modalities.append(img)

        image = torch.tensor(np.stack(modalities, axis=0),
dtype=torch.float32)

        if image.shape[1:] != (96, 96, 96):
            image = F.interpolate(image.unsqueeze(0), size=(96,96,96),
                                  mode='trilinear',
align_corners=False).squeeze(0)

        if self.augment:
            if torch.rand(1) > 0.5: image = torch.flip(image, dims=[1])
            if torch.rand(1) > 0.5: image = torch.flip(image, dims=[2])
            if torch.rand(1) > 0.5: image = torch.flip(image, dims=[3])
            if torch.rand(1) > 0.7:
                image = image + torch.randn_like(image) * 0.01
```

```
        clinical = torch.tensor([row['Sex'], row['Age at MRI']],
dtype=torch.float32)
        label    = torch.tensor(row['MGMT_label'], dtype=torch.long)
        return image, clinical, label

print('✅ Dataset defined (brain-tissue norm + [-5,5] clip)')
```

A.3 Model Architecture

The following snippet shows the core structure of the proposed 3D DenseNet model.

```
class MGMTDenseNet(nn.Module):
    def __init__(self):
        super().__init__()
        self.stem   = nn.Sequential(
            nn.Conv3d(4, 48, 3, padding=1, bias=False),   # ✅ 32→48
stem channels
            nn.BatchNorm3d(48), nn.ReLU(inplace=True), nn.MaxPool3d(2)
        )
        self.block1 = DenseBlock(48,  4, 16)
        self.trans1 = Transition(self.block1.out_channels, 80)
        self.block2 = DenseBlock(80,  4, 16)
        self.trans2 = Transition(self.block2.out_channels, 112)
        self.block3 = DenseBlock(112, 4, 16)

        # ✅ Mixed pooling: avg captures mean signal, max captures peak
activations
        self.avg_pool = nn.AdaptiveAvgPool3d(1)
        self.max_pool = nn.AdaptiveMaxPool3d(1)
        img_feat_dim  = self.block3.out_channels * 2   # concat avg+max

        self.img_fc   = nn.Sequential(nn.Linear(img_feat_dim, 128),
                                      nn.ReLU(), nn.Dropout(0.35))
```

```
        self.clin_mlp = nn.Sequential(nn.Linear(2, 16), nn.ReLU(),
                                      nn.Dropout(0.2), nn.Linear(16,
16))
        self.fusion   = GatedFusion(128, 16, 64)
        self.head     = nn.Sequential(nn.Linear(64, 32), nn.ReLU(),
                                      nn.Dropout(0.25), nn.Linear(32,
2))

    def forward(self, img, clin):
        x = self.stem(img)
        x = self.trans1(self.block1(x))
        x = self.trans2(self.block2(x))
        x = self.block3(x)
        x = torch.cat([self.avg_pool(x).flatten(1),
                       self.max_pool(x).flatten(1)], dim=1)   # mixed
pool
        return self.head(self.fusion(self.img_fc(x),
self.clin_mlp(clin)))
```

A.4 Training Procedure

The following code snippet shows the simplified training loop used in the project.

```
for epoch in range(NUM_EPOCHS):
        model.train()
        correct, total = 0, 0
        for imgs, clin, lbls in train_loader:
            imgs, clin, lbls = imgs.to(DEVICE), clin.to(DEVICE),
lbls.to(DEVICE)
            optimizer.zero_grad()
            out  = model(imgs, clin)
            loss = criterion(out, lbls)
            loss.backward()
```

```
            torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
            optimizer.step()
            correct += (out.argmax(1) == lbls).sum().item()
            total   += lbls.size(0)
        train_acc = correct / total
```

[Github Repository Link]